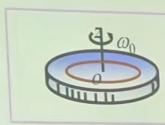


1、绕中心轴以 ω_0 旋转的半径 R_1 和 R_2 的塑料圆环，电荷 q 均匀分布在盘面。求：中心 o 的磁感应强度。



$$R_2 > R_1$$

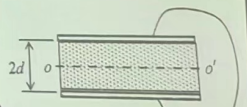
$$dq = \frac{q}{\pi(R_2^2 - R_1^2)} \cdot 2\pi r \cdot dr \quad T = \frac{2\pi}{\omega_0}$$

$$dI = \frac{dq}{T} = \frac{\omega_0 q r}{\pi(R_2^2 - R_1^2)} dr$$

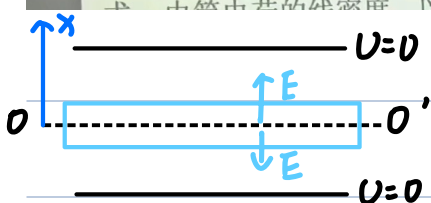
$$dB = \frac{\mu_0 dI}{2r} = \frac{\mu_0 \omega_0 q}{2\pi(R_2^2 - R_1^2)} dr$$

$$B = \int_{R_1}^{R_2} dB = \frac{\mu_0 \omega_0 q}{2\pi(R_2 + R_1)}$$

2、两块接地大导体板间充满均匀密度 n 的电子气。求极板间的场强和电势分布。



3、同轴电容器电势差为 U ， $R_1=2\text{cm}$ ， $R_2=4\text{cm}$



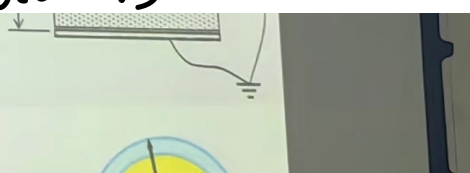
做高斯面，面积 ΔS (对称分布，电场、电势均对称)

$$E \cdot 2\Delta S = \frac{n e \Delta S \cdot 2x}{\epsilon_0} \Rightarrow E = \frac{n e x}{\epsilon_0} \quad (x \text{ 为到 } OO' \text{ 距离})$$

$$\text{电势 } 0 < x < d \quad U = \int_x^d E dx = \frac{n e}{\epsilon_0} (d^2 - x^2)$$

$$-d < x < 0 \quad U = \int_x^{-d} E dx = \frac{n e}{\epsilon_0} (d^2 - x^2)$$

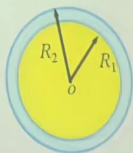
3、同轴电容器电势差为 U ， $R_1=2\text{cm}$ ， $R_2=4\text{cm}$
求：内筒电荷的线密度，以及 $r=3\text{cm}$ 处的电势。



$$U = \int_{R_1}^{R_2} \frac{\lambda}{2\pi\epsilon_0 r} dr = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R_2}{R_1} = \frac{\lambda \ln 2}{2\pi\epsilon_0} \Rightarrow \lambda = \frac{2\pi\epsilon_0 U}{\ln 2}$$

$$U(r=3\text{cm}) = \int_r^{R_2} \frac{\lambda}{2\pi\epsilon_0 r} dr = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{4}{3} = \frac{U}{\ln 2} \ln \frac{4}{3}$$

4、半径 R_1 介质球，电荷体密度 $\rho(r)=\rho_0 r$ ，相对电容率 $\epsilon_r=2$ ，外包半径为 R_2 的介质层， $\epsilon_r=5$ 。求



- 1) 电场强度和电势的分布；
- 2) 半径 R_1 界面处极化电荷和总静电能。

$$1) \text{ 场强: } ① \quad r < R_1 \quad \oint E_1 dS = \frac{\Sigma q}{\epsilon} = \frac{\int_0^r \rho(r) \cdot 4\pi r^2 dr}{\epsilon_1 \epsilon_0} = \frac{\rho_0 \pi r^4}{2\epsilon_0} = E_1 \cdot 4\pi r^2$$

$$E_1 = \frac{\rho_0 r^2}{8\epsilon_0}$$

$$② \quad R_1 < r < R_2 \quad \oint E_2 dS = \frac{\Sigma q}{\epsilon} = \frac{\int_0^{R_1} \rho(r) \cdot 4\pi r^2 dr}{\epsilon_r \epsilon_0} = \frac{\rho_0 \pi R_1^4}{5\epsilon_0} = E_2 \cdot 4\pi r^2$$

$$E_2 = \frac{\rho_0 R_1^4}{20\epsilon_0 r^2}$$

$$\textcircled{3} \quad r > R_2 \quad \oint E_3 ds = \frac{\Sigma q}{\epsilon_0} = \frac{\rho_0 \pi R_1^4}{\epsilon_0} = E_3 \cdot 4\pi r^2$$

$$E_3 = \frac{\rho_0 R_1^4}{4\epsilon_0 r^2}$$

电势: (考试从 U_1 开始的后面发现要先算 U_3, U_2)

$$U_3(r) = \int_r^\infty E_3 dr = \frac{\rho_0 R_1^4}{4\epsilon_0 r}$$

$$U_2(r) = \int_r^{R_2} E_2 dr + U_3(R_2)$$

$$= \frac{\rho_0 R_1^4}{20\epsilon_0} \left(\frac{1}{r} - \frac{1}{R_2} \right) + \frac{\rho_0 R_1^4}{4\epsilon_0 R_2} = \frac{\rho_0 R_1^4}{20\epsilon_0 r} + \frac{\rho_0 R_1^4}{5\epsilon_0 R_2}$$

$$U_1(r) = \int_r^{R_1} E_1 dr + U_2(R_1)$$

$$= \frac{\rho_0}{24\epsilon_0} (R_1^3 - r^3) + \frac{\rho_0 R_1^3}{20\epsilon_0} + \frac{\rho_0 R_1^4}{5\epsilon_0 R_2}$$

2) R_1 处 $\sigma_p = P_1 - P_2$

$$= (\epsilon_1 - 1)\epsilon_0 E_1 - (\epsilon_r - 1)\epsilon_0 E_2$$

$$= \epsilon_0 \cdot \frac{\rho_0 R_1^2}{8\epsilon_0} - 4\epsilon_0 \frac{\rho_0 R_1^4}{20\epsilon_0 R_2^2}$$

$$= -\frac{3\rho_0 R_1^2}{40}$$

$$\Sigma q' = S \sigma_p = 4\pi R_1^2 \sigma_p = -\frac{3\rho_0 R_1^4 \pi}{10}$$

$$W = \int \frac{1}{2} \epsilon E^2 dV$$

$$= \int \frac{1}{2} \epsilon_1 E_1^2 \cdot 4\pi r^2 dr + \int \frac{1}{2} \epsilon_r E_2^2 \cdot 4\pi r^2 dr + \int \frac{1}{2} \epsilon_0 E_3^2 \cdot 4\pi r^2 dr$$

不想算了 $\pi\pi$, - 着急算错了